PIZZA SALES ANALYSIS

SQL Query Report & Business Insights

Problem Statement

A pizza restaurant chain holds transactional sales data covering every order placed across its locations. While the raw data captures individual line items, the business lacks a clear, structured view of its own performance.

Key commercial questions: how much revenue is being generated, which days drive the most demand, and how efficiently customers are spending, remain unanswered without targeted analysis.

This project uses SQL to interrogate the dataset and surface the metrics that matter most to the business. Eight queries are developed, ranging from high-level revenue aggregates to time-series breakdowns by day and month. Each query is designed to answer a specific business question, with results that can feed directly into dashboards, operational decisions and reporting.

The analysis demonstrates core SQL competencies including aggregate functions, distinct counting, type casting for precision and date-part extraction, all applied in a realistic business context.

Tools & Dataset

Language: SQL

Dataset: pizza_sales_excel_file — transactional pizza order records

Key fields: order_id, order_date, quantity, total_price

Scope: 8 queries covering revenue, volume, efficiency and time-series metrics

SQL Queries & Analysis

QUERY 01

Total Revenue

SQL Query

```
SELECT SUM(total_price) AS Total_Revenue
FROM pizza_sales_excel_file
```

Query Result

Results | Messages

	Total_Revenue
1	817860.05083847

> **Total revenue is the headline metric, giving stakeholders an immediate view of overall performance. is the starting point for profitability analysis and is best tracked over time to identify seasonal peaks and troughs. A single This figure aggregate like this also acts as a sanity check on data quality — any anomalous figure here signals an upstream data issue worth investigating before deeper analysis is carried out.**

Metric type
Aggregate SUM

Business use
P&L reporting

QUERY 02

Average Order Value

SQL Query

```
SELECT SUM(total_price) / COUNT(DISTINCT order_id)
  AS Average_Order_Value
FROM pizza_sales_excel_file
```

Query Result

Results | Messages

	Avergage_order_value
1	38.3072623343546

> **Average order value (AOV) is a critical revenue efficiency metric. A rising AOV suggests customers are either ordering more items per transaction. Tracking AOV alongside total orders reveals whether revenue growth is driven by volume or by higher spend per customer, which informs very different marketing and menu strategies.**

QUERY 03

Total Pizzas Sold

SQL Query

```
SELECT SUM(quantity) AS Total_Pizza_Sold
FROM pizza_sales_excel_file
```

Query Result

Results | Messages

	Total_Pizza_Sold
1	49574

Total pizzas sold is a volume metric that complements revenue figures. A high volume with lower-than-expected revenue can indicate over-reliance on discounting or a bias towards lower-priced menu items. Conversely, lower volume with strong revenue points to premium product preference. This metric is also essential for supply chain and inventory planning — understanding how many units move over a given period directly informs purchasing and staffing decisions.

Metric type
Volume count

Aggregation
SUM(quantity)

QUERY 04

Total Orders

SQL Query

```
SELECT COUNT(DISTINCT order_id) AS Total_Ordes
FROM pizza_sales_excel_file
```

Query Result

Results | Messag

	Total_ordes
1	21350

Total distinct orders measures customer transaction frequency. Using COUNT(DISTINCT order_id) avoids double-counting multi-item orders. Comparing total orders against total pizzas sold yields average pizzas per order, enabling group-ordering behaviour to be identified and targeted with promotional strategies.

QUERY 05

Average Pizzas Per Order

SQL Query

```
SELECT SUM(quantity) / COUNT(DISTINCT order_id)
  AS Average_Pizza_Per_Order
FROM pizza_sales_excel_file
```

Query Result

Results | Messages

	Average_pizza_per_order
1	2

This metric show how many pizzas on average are ordered per order. A higher average suggests customers are frequently ordering for multiple people. The integer division version of this query gives a quick estimate, while the CAST-based decimal query (KPI 06) provides the precise figure. Understanding basket size is essential for upsell targeting.

Formula
Units ÷ Orders

Metric type
Behavioural KPI

QUERY 06

Pizzas Per Order (Precise)

SQL Query

```
SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2))
  / CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2))
  AS DECIMAL(10,2)) AS Pizza_Per_Order
FROM pizza_sales_excel_file
```

Query Result

Results | Messages

	pizza_per_order
1	2.32

This refined version of KPI 05 uses explicit DECIMAL casting to avoid integer truncation, producing a precise two-decimal-place result. This is an important demonstration of SQL rigour — integer division silently truncates results, which can mislead business stakeholders if not corrected.

QUERY 07

Total Orders by Day of Week

SQL Query

```
SELECT DATENAME(DW, order_date) AS Order_Day,
       COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales_excel_file
GROUP BY DATENAME(DW, order_date)
```

Output Chart

Results | Messages

	order_day	Total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

Order distribution by day of week is one of the most actionable insights in any food service dataset. Peak days typically fall on Fridays and weekends, reflecting social dining patterns. Identifying the slowest trading day can enable targeted promotions.

Function used
DATENAME(DW)

Grouping
By weekday

QUERY 08

Total Orders by Month

SQL Query

```
SELECT DATENAME(MONTH, order_date) AS Order_Month,
       COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales_excel_file
GROUP BY DATENAME(MONTH, order_date)
```

Output Chart

Month_name	Total_orders
July	1935
May	1853
January	1845
August	1841
March	1840
April	1799
November	1792
June	1773
February	1685
December	1680
September	1661
October	1646

Monthly order trends reveal seasonality patterns that are invisible in aggregate figures. Peak months often align with summer (outdoor gatherings, school holidays) or festive periods, while quieter months present opportunities for targeted campaigns. This time-series view is essential for accurate revenue forecasting and inventory management.

Function used
DATENAME(MONTH)

Grouping
By month

Summary & Conclusions

1. Revenue and order volume metrics confirm the business is generating consistent sales but the real value lies in combining these figures — AOV and pizzas-per-order reveal whether growth is driven by volume or by higher spend per customer.
2. The precise DECIMAL casting in KPI 06 versus the integer version in KPI 05 illustrates an important SQL discipline: always validate data types before performing division to avoid silent truncation errors in production reports.
3. Day-of-week analysis enables targeted staffing and promotional decisions, reducing labour costs on slow days and ensuring adequate coverage during peak periods.
4. Monthly seasonal patterns provide the foundation for accurate revenue forecasting and inventory planning, allowing the business to avoid both stockouts and excess waste.

Together, these eight queries form a complete analytical foundation, moving from raw transactional data to business-ready KPIs that can feed directly into a live dashboard or executive report.